



Analysing Contact Learning Hours in UK FE and Sixth Form Colleges

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Contents	
1. Summary	2
2. Methods.....	2
2.1. Data Preprocessing	2
2.2. Data Summaries.....	3
2.3. Exploratory Data Analysis	4
2.4 Choice of Statistical Model	5
3. Results and Interpretation.....	5
3.1 NPAR1WAY	5
3.2. ROBUSTREG	6
4. Conclusions	7
5. Appendix	8
Appendix A: File Names and Metadata	8
Appendix B: Methods	9
Appendix C: Box Plots for Outlier Analysis	12
Appendix D: Transformation	14
Appendix E: NPAR1WAY	15
6. References and Bibliography	26

1. Summary

The report examines the impact of region, institution type, size and year on the number of Contact Learning Hours (CH) per learner of sixth form and FE institutions in the UK. In comparing with FE institutions, the study found that sixth form colleges consistently outperform their higher education colleagues in CH per learner and other performance indicators. The evidence that the large educational institutions perform better than smaller ones suggests that the scale of the institution influences the student performance. VAF, which has a positive effect on CH per learner, indicates an importance of better resources and facilities in achievement. These findings emphasise the importance of resource allocations and making CH_L a key measure for assessing the college's effectiveness, providing valuable insights for future educational policies and practices.

2. Methods

2.1. Data Preprocessing

- **CSIP5302-6form.csv** - The file was imported using the FILENAME along with INFILE and INPUT. (See Appendix A Fig 2.1.1.)
- **CSIP5302-FEmetric02.tab** - The PROC IMPORT was utilised for this file (See Appendix A Fig 2.1.2).
- **CSIP5302-FE.csv** - A MACRO has been used to read the data file and paired with INFILE to import the file (See Appendix A Fig 2.1.3).
- By utilizing PROC MEANS (See Appendix A Fig 2.1.4) and SQL, average of numeric variables was calculated, and missing values were replaced with these computed values for consistency of data.
- Rows that were unnecessary or not in-line with the organization of the dataset were removed like rows containing subtotal and grand-total values.
- PROC SQL was used to join FE and FE_Metrics by ID (See Appendix A Fig 2.1.6.)

Education Institutions Data										
Obs	GradPR	VAF	InstitutionType	Region	GPercentFemale	GPercentMale	CH_per_Learner1	CH_per_Learner2	CH_per_Learner3	CH_per_Learner
1	0.91	0.57	Sixth Form College	East Midlands	50	50	721.386	737.229	729.519	729.378
2	0.9	0.45	Sixth Form College	East Midlands	50	50	755.172	772.789	569.814	699.252
3	0.82	0.53	Sixth Form College	East Midlands	49	51	516.060	537.753	449.634	501.149
4	0.85	0.5	Sixth Form College	East Midlands	51	49	683.631	706.171	683.909	691.237
5	0.84	0.49	Sixth Form College	East of England	50	50	576.012	564.667	448.725	529.801
6	0.83	0.58	Sixth Form College	East of England	49	51	220.525	220.175	245.774	228.825
7	0.79	0.54	Sixth Form College	East of England	49	51	522.824	602.682	544.666	556.724
8	0.75	0.49	Sixth Form College	East of England	48	52	652.424	661.491	673.876	662.597
9	0.81	0.49	Sixth Form College	East of England	47	53	709.517	726.301	606.826	680.882
10	0.74	0.52	Sixth Form College	East of England	50	50	601.566	379.509	307.890	429.655

Fig 2.1.7 Appended 6Form and FE

344	0.726991	0.356427	FE College	Yorkshire and the Humber	52	48	180.565	217.113	226.593	208.091
345	0.8398404	0.3139215	FE College	Yorkshire and the Humber	52	48	129.330	169.594	126.415	141.780
346	0.7387261	0.3385849	FE College	Yorkshire and the Humber	51	49	180.599	188.322	150.803	173.241
347	0.7705272	0.3250564	FE College	Yorkshire and the Humber	51	49	77.660	86.580	104.022	89.421
348	0.7769626	0.3315369	FE College	Yorkshire and the Humber	51	49	98.498	155.201	124.573	126.091
349	0.8412267	0.3970145	FE College	Yorkshire and the Humber	49	51	146.875	154.985	141.501	147.787
350	0.7895996	0.3554396	FE College	Yorkshire and the Humber	48	52	113.755	131.675	120.086	121.838
351	0.7252454	0.372095	FE College	Yorkshire and the Humber	52	48	184.174	197.277	193.902	191.784
352	0.7095748	0.3988824	FE College	Yorkshire and the Humber	48	52	204.967	230.795	216.582	217.448
353	0.7311053	0.3321484	FE College	Yorkshire and the Humber	52	48	269.456	312.437	286.150	289.348
354	0.7695804	0.3331197	FE College	Yorkshire and the Humber	49	51	125.748	205.799	202.218	177.921

Fig 2.1.8 Appended 6Form and FE

2.2. Data Summaries

Descriptive Statistics of Education Institutions Data

The MEANS Procedure

Variable	Mean	Std Dev	Skewness	Kurtosis	N	N Miss
GPercentFemale	49.9184699	1.4871175	-0.2129814	-0.9756181	354	0
GPercentMale	50.0815301	1.4871175	0.2129814	-0.9756181	354	0
GradPR	0.7802018	0.0681014	0.2621689	-0.3328921	354	0
VAF	0.4049044	0.0994996	0.6333341	-0.7576953	354	0
CH_per_Learner1	285.7000765	212.0989444	1.2488035	0.2271596	354	0
CH_per_Learner2	305.2767247	194.0556987	1.3153398	0.4900750	354	0
CH_per_Learner3	290.2899232	183.4900470	1.3494476	0.7149296	354	0
CH_per_Learner	293.7555748	194.4618704	1.3080073	0.4592808	354	0

Fig 2.2.1 The above figure shows the Cleaned dataset which has no missing values

- PROC CONTENTS (See Appendix A Fig 2.2.2 & 2.2.3) is used to provide a detailed overview of the structure of the dataset.
- Analysis of PROC TABULATE (Carpenter) is discussed below.
 - There is a lot of variety, the Southwest has a heavy-tailed distribution while Greater London has the highest mean in CH_L.
 - Southeast has the highest VAF, Yorkshire and the Humber has the highest graduation%.
 - Sixth-Form Colleges perform consistently better in all factors compared to FE Colleges
 - Larger Institutions offer more CH_L and have higher graduation% than smaller institutions.
 - Gender Ratios are generally balanced across regions with minor-differences among smaller institutions and Sixth-Form Colleges.

Statistical summary of Region, InstitutionType, InstitutionSize Vs GradPR, VAF, GPercentFemale, GPercentMale, CH_per_Learner

	GradPR				VAF				GPercentFemale				GPercentMale				CH_per_Learner			
	Mean	Std	Skewness	Kurtosis	Mean	Std	Skewness	Kurtosis	Mean	Std	Skewness	Kurtosis	Mean	Std	Skewness	Kurtosis	Mean	Std	Skewness	Kurtosis
Region																				
East Midlands	0.77	0.07	0.51	-0.35	0.39	0.08	0.58	-0.11	50.33	1.24	-0.40	-0.86	49.67	1.24	0.40	-0.86	245.74	195.21	1.88	2.13
East of England	0.76	0.05	-0.72	0.65	0.39	0.10	0.48	-1.14	49.77	1.31	0.07	-0.43	50.23	1.31	-0.07	-0.43	305.10	176.31	1.50	1.46
Greater London	0.78	0.07	0.21	-0.61	0.40	0.11	0.80	-0.47	49.83	1.54	-0.15	-1.06	50.17	1.54	0.15	-1.06	335.49	213.97	1.20	-0.03
North East	0.77	0.07	0.07	-0.35	0.41	0.10	0.51	-1.04	50.00	1.26	0.16	-1.08	50.00	1.26	-0.16	-1.08	286.73	214.24	1.05	-0.69
North West	0.78	0.06	0.18	-0.79	0.41	0.10	0.46	-0.94	49.46	1.59	0.10	-1.11	50.54	1.59	-0.10	-1.11	333.35	216.45	1.05	-0.36
South East	0.79	0.07	0.06	-0.94	0.43	0.10	0.31	-1.34	50.27	1.36	-0.39	-0.68	49.73	1.36	0.39	-0.68	293.98	168.78	1.35	1.54
South West	0.77	0.05	0.63	-0.13	0.35	0.08	1.79	3.81	50.17	1.44	-0.17	-1.28	49.83	1.44	0.17	-1.28	212.37	140.73	3.04	9.69
West Midlands	0.78	0.07	-0.07	-0.80	0.41	0.09	0.72	-0.60	49.89	1.43	-0.18	-0.98	50.11	1.43	0.18	-0.98	264.64	176.81	1.31	0.31
Yorkshire and the Humber	0.80	0.08	0.62	0.06	0.41	0.11	0.83	-0.63	49.87	1.81	-0.36	-1.27	50.13	1.81	0.36	-1.27	299.93	214.29	1.18	-0.04
InstitutionType																				
FE College	0.75	0.05	0.00	-0.28	0.35	0.05	0.08	-0.31	50.14	1.32	-0.12	-1.12	49.86	1.32	0.12	-1.12	190.92	59.16	0.81	0.74
Sixth Form College	0.85	0.06	-0.53	0.75	0.54	0.05	-0.45	-0.05	49.36	1.72	0.08	-1.27	50.64	1.72	-0.08	-1.27	554.96	172.75	-0.43	-0.69
InstitutionSize																				
Large	0.84	0.06	-0.77	0.62	0.55	0.05	-0.18	-1.02	48.89	1.68	0.51	-1.03	51.11	1.68	-0.51	-1.03	731.29	53.48	0.87	-0.42
Large-Medium	0.83	0.06	-0.38	0.79	0.52	0.08	-1.82	3.63	49.68	1.67	-0.18	-1.12	50.32	1.67	0.18	-1.12	509.73	94.34	-0.08	-1.25
Medium	0.77	0.06	0.27	-0.40	0.38	0.08	1.02	0.79	49.74	1.32	0.01	-0.68	50.26	1.32	-0.01	-0.68	260.38	40.52	0.66	-0.52
Small	0.76	0.06	0.16	-0.18	0.36	0.06	1.06	2.10	50.20	1.35	-0.32	-1.13	49.80	1.35	0.32	-1.13	133.59	21.56	-0.69	-0.30
Small-Medium	0.75	0.05	0.10	0.25	0.35	0.05	0.77	2.26	50.37	1.35	-0.24	-1.16	49.63	1.35	0.24	-1.16	185.40	13.32	0.18	-1.21

Fig 2.2.4 PROC TABULATE

2.3. Exploratory Data Analysis

- (Han et al., 2022) states that when the graph of points inclines from top-left to bottom-right, meaning that there is a negative correlation.

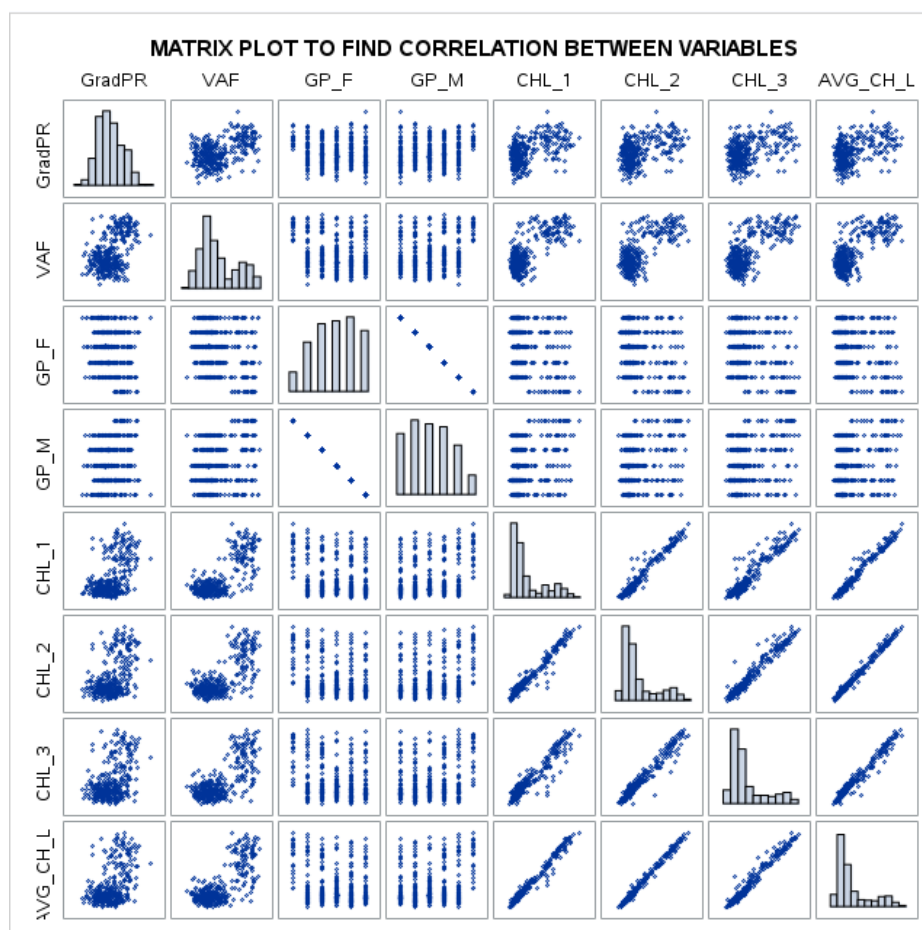


Fig 2.3.1 The above figure shows positive correlation between CHL_1, CHL_2, CHL_3 with AVG_CH_L and negative correlation between GP_F and GP_M

- As the variable GPercentMale has a positive influence on the data, we use this for the analysis and GPercentFemale analysis will be implied automatically.
- As CH_per_Learner 1,2,3 are the only variables that vary with years, it is understood that the analysis done on the average will be true for 3 years of data. Hence, the analysis will be done only on CH_per_Learner.
- A box plot was utilized in identifying outliers in the dataset (See Appendix C). As outliers were detected in CH_per_Learner, which is our primary variable of analysis, it will be converted to normalize its distribution and minimize the impact of outliers.
- Box-Cox transformation is employed to determine the optimal shape of transformation to correct outliers for nonnegative response variables. The transformation is λ -parameterized, with different values referring to varying transformations (Atkinson et al., 2021) (See Appendix D Fig 2.3.2).

2.4 Choice of Statistical Model

As the outliers persist (See Appendix D Fig 2.4.1 & 2.4.2), we choose models that handle them and do analysis accordingly.

NPARIWAY is a nonparametric method, the tests involved here do not depend on normal distribution of data but instead work with ranked data (Padgett, 2011). **Robust Regression** is another method which is implemented to detect outliers and to provide stable results in the presence of outliers (Chen, 2002).

3. Results and Interpretations

3.1 NPARIWAY

- Region (See Appendix E 3.1.1)
 - Differences are notable, with Southeast having higher mean scores than other regions such as Southwest, suggesting higher VAF for pupils in some parts.

- Strongly significant differences with highest mean score of Greater London and lowest for Southwest
- Institution Type (See Appendix E 3.1.2)
 - Sixth Form Colleges have higher graduation% than FE Colleges whose higher rank scores and mean values affirm this.
 - Sixth Form Colleges have much higher rank scores and mean values, which affirm their value in adding value to learners' development.
 - Sixth Form Colleges provide far more CH compared to FE Colleges
- Institution Size (See Appendix E 3.1.3)
 - The largest institutions provide the highest mean scores, with lesser institutions providing increasingly lower GradPR.
 - The larger education institutions have considerably higher VAF mean scores.

Statistical Modelling By Region - NPAR1WAY

The NPAR1WAY Procedure

Region	N	Sum of Scores	Expected Under H0	Std Dev Under H0	Mean Score
East Midlands	24	3183.0	4260.00	484.045452	131.791667
East of England	31	6234.0	5502.50	544.259053	201.096774
Greater London	47	9984.0	8342.50	653.343644	212.425532
North East	21	3197.0	3727.50	454.836509	152.238095
North West	61	12083.0	10827.50	727.147108	198.081967
South East	59	11199.0	10472.50	717.563876	189.813559
South West	29	3754.0	5147.50	528.036852	129.448276
West Midlands	45	6909.0	7987.50	641.370603	153.533333
Yorkshire and the Humber	37	6312.0	6567.50	589.052558	170.594595

Chi-Square	DF	Pr > ChiSq
25.5425	8	0.0013

Fig 3.1.4 Region Vs CH_per_Learner

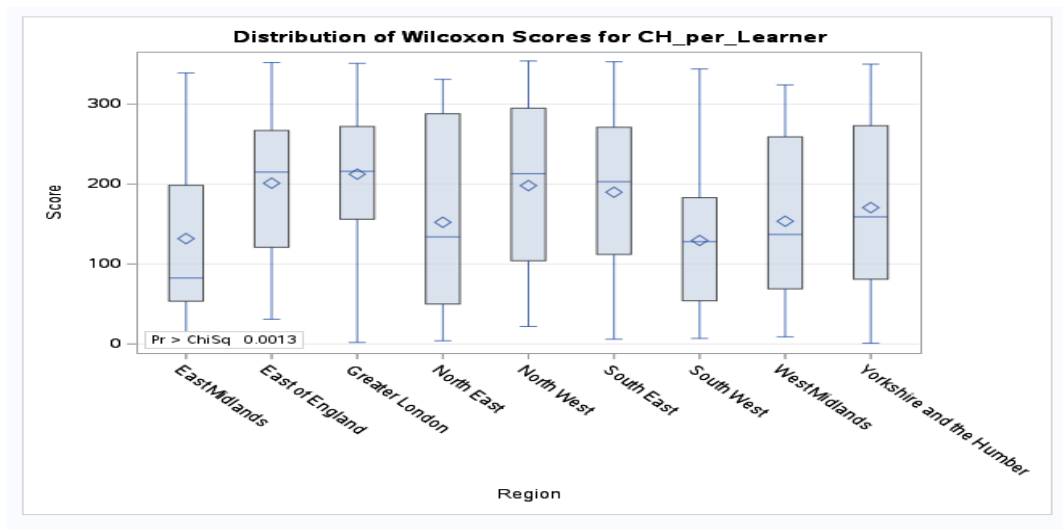


Fig 3.1.5 Distribution of contacts hours across regions

3.2. ROBUSTREG

Summary Statistics						
Variable	Q1	Median	Q3	Mean	Standard Deviation	MAD
VAF	0.3315	0.3713	0.4900	0.4049	0.0995	0.0830
GPercentMale	49.0000	50.0000	51.0000	50.0815	1.4871	1.4826
GradPR	0.7311	0.7715	0.8264	0.7802	0.0661	0.0684
CH_per_Learn	164.4	209.5	351.3	293.8	194.5	94.9866

Fig 3.2.1 Summary Statistics

- **VAF** has a mean of 0.4049 with moderate variability.
- **GPercentMale**: Moderately well-balanced, with a mean of 50.08% and low variability.
- **GradPR**: High Graduation%, with a mean of 0.7802 and minimal variability.
- **CH_per_Learner**: Wide range with outliers and mean of 293.8.

Parameter Estimates							
Parameter	DF	Estimate	Standard Error	95% Confidence Limits		Chi-Square	Pr > ChiSq
Intercept	1	-1083.98	228.6292	-1532.09	-635.876	22.48	<.0001
VAF	1	1277.143	76.5250	1127.157	1427.129	278.53	<.0001
GPercentMale	1	9.4161	4.5218	0.5534	18.2787	4.34	0.0373
GradPR	1	495.0310	113.4933	272.5882	717.4738	19.02	<.0001
Scale	1	112.6978					

Diagnostics Summary		
Observation Type	Proportion	Cutoff
Outlier	0.0113	3.0000

Goodness-of-Fit	
Statistic	Value
R-Square	0.3922
AICR	362.8308
BICR	379.6395
Deviance	4523543

Fig 3.2.2 Parameter Estimates

- The intercept of -1083.98 is the starting point when predictors are zero.
- **VAF**: Represents the greatest positive influence indicating its strong contribution to the outcome variable.
- **GPercentMale**: Has a moderate positive influence indicating a moderate relationship.
- **GradPR**: Indicates a large positive influence which shows its substantial impact on the outcome.

The model represents powerful VAF and GradPR with moderate explanatory variance. Although CH_per_Learner reflects significant variation due to the presence of outliers, the minority proportion of the outliers ensures trivial distortion of analysis.

4. Conclusions

This analysis provides grounds for a consideration of the effect of institution size and type on the learning outcome. Sixth-form colleges and FE colleges have higher contact learning hours (CH) per student, higher value add factors (VAF), and higher graduation rates because of their greater resources and scope for involvement of

the students. There are regional variations, with the Southeast and Greater London having the highest CH and VAF, reflecting differences in the education system in the UK.

Where tiny gender differences capture the need for collective action to attain equity, the positive impact of VAF on CH per pupil underscores the value of infrastructure and resource availability in enabling one to attain superior results. Overall, the findings stress the necessity to strengthen institutional resources and leverage CH_L as a main basis for evaluation.

5. Appendix

Appendix A: File Names and Metadata

1. 6form.csv
2. FEmetric02.tab
3. FE.csv
4. contact_hours_analysis.sas

UK Educational Institutions Data

Variable Name	Data Type	Observations	Anomalies
GradPR	Numeric		
VAF	Numeric		
Institution Type	Character		
Region	Character		
GPercentFemale	Numeric	Present in Sixth Form	Calculated in FE College
GPercentMale	Numeric	Present in FE College	Calculated in Sixth Form
CH_per_Learner1	Numeric		Obtained from Total_CH_Year1/Learners1
CH_per_Learner2	Numeric		Obtained from Total_CH_Year2/Learners2
CH_per_Learner3	Numeric		Obtained from Total_CH_Year3/Learners3
CH_per_Learner	Numeric		Obtained from Average of CH_per_Learners 1,2,3
Institution Size	Character		Calculated based on CH_per_Learner

Appendix B: Methods

SIXTH FORM COLLEGE											
Obs	Total_CH_Year1	Learners1	Total_CH_Year2	Learners2	Total_CH_Year3	Learners3	GradPR	VAF	InstitutionType	Region	GPercentFemale
1	1248720	1731	1115427	1513	1003088	1375	0.91	0.57	Sixth Form College	East Midlands	50
2	972662	1288	995326	1288	1010281	1773	0.90	0.45	Sixth Form College	East Midlands	50
3	857176	1661	867934	1614	815186	1813	0.82	0.53	Sixth Form College	East Midlands	49
4	1412381	2066	1315597	1863	1291220	1888	0.85	0.50	Sixth Form College	East Midlands	51
5	4490939	6746	4294284	6278	4119775	6849	.	.	Sub Total East Midlands		.
6	900882	1564	819897	1452	772705	1722	0.84	0.49	Sixth Form College	East of England	50
7	1168121	5297	1181678	5367	1175536	4783	0.83	0.58	Sixth Form College	East of England	49
8	1116752	2136	1297574	2153	1458615	2678	0.79	0.54	Sixth Form College	East of England	49
9	1416412	2171	1330920	2012	1323492	1964	0.75	0.49	Sixth Form College	East of England	48
10	1446706	2039	1428635	1967	1257344	2072	0.81	0.49	Sixth Form College	East of England	47

Fig 2.1.1: Sixth Form Data

FE METRICS			
Obs	ID	GradPR	VAF
1	1	0.7321957	0.3100195
2	2	0.8312862	0.2676687
3	3	0.7687803	0.4189092
4	4	0.7831351	0.4704178
5	5	0.7666574	0.4295499
6	6	0.8492783	0.3469985
7	7	0.7338743	0.3546149
8	8	0.6926224	0.3344194
9	9	0.6722447	0.3350848
10	10	0.769926	0.3645743

Fig 2.1.2: FE Metrics

FE COLLEGE										
Obs	ID	InstitutionType	Region	Total_CH_Year1	Learners1	Total_CH_Year2	Learners2	Total_CH_Year3	Learners3	GPercentMale
1	1	FE College	East Midlands	1299123	9919	1294331	7400	1107795	6630	49
2	2	FE College	East Midlands	1593307	14459	2701354	16526	2543711	14930	48
3	3	FE College	East Midlands	3208343	20923	3088582	12716	2743416	11715	50
4	4	FE College	East Midlands	2615180	36433	3132412	30146	3027782	28989	49
5	5	FE College	East Midlands	1015354	7449	933109	5445	890615	5214	51
6	6	FE College	East Midlands	3713574	27503	3625550	23288	3246155	22447	49
7	7	FE College	East Midlands	2258863	10548	2226036	9825	2648944	13113	49
8	8	FE College	East Midlands	1731174	10303	1841059	9827	1977806	9863	48
9	9	FE College	East Midlands	1006989	4042	1013832	3584	1010010	3725	48
10	10	FE College	East Midlands	4402042	20514	4243906	17834	3739237	17359	50

Fig 2.1.3: FE College

Descriptive Statistics of SIXTH FORM COLLEGE

The MEANS Procedure

Variable	Mean	Std Dev	Skewness	Kurtosis	N	N Miss
Total_CH_Year1	3104689.64	11170827.54	8.6166431	80.8991106	106	4
Learners1	5860.27	21273.53	8.4577434	78.1940843	106	4
Total_CH_Year2	3015470.00	10963429.73	8.6843073	82.2107430	108	2
Learners2	6034.42	22203.78	8.4729510	78.5599696	108	2
Total_CH_Year3	2916460.99	10705373.74	8.7628488	83.7287436	110	0
Learners3	6510.87	24126.87	8.5829638	80.7271182	110	0
GradPR	0.8459406	0.0554288	-0.5348908	0.7868807	101	9
VAF	0.5439604	0.0463482	-0.4474292	-0.0199288	101	9
GPercentFemale	49.3600000	1.7204651	0.0786331	-1.2747087	100	10

Descriptive Statistics of FE METRICS

The MEANS Procedure

Variable	Mean	Std Dev	Skewness	Kurtosis	N	N Miss
ID	128.0000000	73.7563557	0	-1.2000000	255	1
GradPR	0.7543364	0.0501436	0.0039208	-0.2926413	253	3
VAF	0.3501424	0.0481189	0.0789186	-0.3188852	253	3

Descriptive Statistics of FE COLLEGE

The MEANS Procedure

Variable	Mean	Std Dev	Skewness	Kurtosis	N	N Miss
ID	128.0000000	73.7563557	0	-1.2000000	255	1
Total_CH_Year1	2414477.16	5387276.77	12.3901161	167.1492020	254	2
Learners1	14840.59	29684.15	10.7564483	123.7921638	254	2
Total_CH_Year2	2482533.15	5420859.25	12.1727905	161.9580845	251	5
Learners2	12672.90	26126.44	10.9602976	130.8944544	251	5
Total_CH_Year3	2309349.66	5081726.78	11.9470030	156.2741040	245	11
Learners3	12170.67	24718.59	10.6202471	123.3792678	245	11
GPercentMale	49.8616601	1.3276039	0.1230832	-1.1290557	253	3

Fig 2.1.4: Descriptive Statistics of Datasets

Descriptive Statistics of Cleaned SIXTH FORM COLLEGE

The MEANS Procedure

Variable	Mean	Std Dev	Skewness	Kurtosis	N	N Miss
Total_CH_Year1	1221177.94	554838.73	1.8635976	3.7531071	100	0
Learners1	2305.04	1276.39	1.6549759	2.4487818	100	0
Total_CH_Year2	1145878.60	472377.67	1.4598164	4.0831833	100	0
Learners2	2293.08	1346.59	2.0810688	5.0026847	100	0
Total_CH_Year3	1069369.03	373622.38	0.6804752	0.9825784	100	0
Learners3	2387.32	1518.96	2.4282864	6.9327990	100	0
GradPR	0.8459000	0.0557066	-0.5301337	0.7482698	100	0
VAF	0.5440000	0.0485800	-0.4479220	-0.0477298	100	0
GPercentFemale	49.3600000	1.7204651	0.0786331	-1.2747087	100	0
GPercentMale	50.6400000	1.7204651	-0.0786331	-1.2747087	100	0

Descriptive Statistics of Cleaned FE METRICS

The MEANS Procedure

Variable	Mean	Std Dev	Skewness	Kurtosis	N	N Miss
GradPR	0.7543364	0.0499458	0.0039362	-0.2711001	255	0
VAF	0.3501424	0.0479291	0.0792264	-0.2975473	255	0

Descriptive Statistics of Cleaned FE COLLEGE

The MEANS Procedure

Variable	Mean	Std Dev	Skewness	Kurtosis	N	N Miss
Total_CH_Year1	1977900.15	1035601.32	0.9168576	0.8587926	254	0
Learners1	12327.99	6837.93	1.6254825	3.8032629	254	0
Total_CH_Year2	2045885.80	1102063.21	1.3669581	3.5364680	254	0
Learners2	10508.71	6297.51	2.2857913	8.7438007	254	0
Total_CH_Year3	1904878.38	1051698.10	1.5796227	5.2244160	254	0
Learners3	10151.89	6309.72	2.6419653	11.7175175	254	0
GPercentMale	49.8616601	1.3249776	0.1233233	-1.1215246	254	0

Fig 2.1.5: Descriptive Statistics of Cleaned Datasets

FE College with Metrics Data

Obs	ID	InstitutionType	Region	Total_CH_Year1	Learners1	Total_CH_Year2	Learners2	Total_CH_Year3	Learners3	GPercentMale	GPercentFemale	GradPR	VAF
1	1	FE College	East Midlands	1299123	9919	1294331	7400	1107795	6630	49	51	0.7321957	0.3100195
2	2	FE College	East Midlands	1593307	14459	2701354	16526	2543711	14930	48	52	0.8312862	0.2676687
3	3	FE College	East Midlands	3208343	20923	3088582	12716	2743416	11715	50	50	0.7687803	0.4189092
4	4	FE College	East Midlands	2615180	36433	3132412	30146	3027782	28989	49	51	0.7831351	0.4704178
5	5	FE College	East Midlands	1015354	7449	933109	5445	890615	5214	51	49	0.7666574	0.4295499
6	6	FE College	East Midlands	3713574	27503	3625550	23288	3246155	22447	49	51	0.8492783	0.3469985
7	7	FE College	East Midlands	2258893	10548	2226036	9825	2648944	13113	49	51	0.7338743	0.3546149
8	8	FE College	East Midlands	1731174	10303	1841059	9827	1977806	9883	48	52	0.6926224	0.3344194
9	9	FE College	East Midlands	1006969	4042	1013632	3584	1010010	3725	48	52	0.6722447	0.3350848
10	10	FE College	East Midlands	4402042	20514	4243906	17834	3739237	17359	50	50	0.769926	0.3645743

Fig 2.1.6: Further Education College Combined with Metrics Data

The CONTENTS Procedure

Data Set Name	P2911159.EDUCATION_INSTITUTIONS_DATA	Observations	354
Member Type	DATA	Variables	11
Engine	V9	Indexes	0
Created	03/23/2025 14:06:00	Observation Length	184
Last Modified	03/23/2025 14:06:00	Deleted Observations	0
Protection		Compressed	NO
Data Set Type		Sorted	NO
Label			
Data Representation	SOLARIS_X86_64, LINUX_X86_64, ALPHA_TRU64, LINUX_IA64		
Encoding	utf-8 Unicode (UTF-8)		

Engine/Host Dependent Information	
Data Set Page Size	131072
Number of Data Set Pages	1
First Data Page	1
Max Obs per Page	711
Obs in First Data Page	354
Number of Data Set Repairs	0
Filename	/home/u64180336/P2911159/education_institutions_data.sas7bdat
Release Created	9.0401M7
Host Created	Linux
Inode Number	5914891719
Access Permission	rw-r--r--
Owner Name	u64180336
File Size	256KB
File Size (bytes)	262144

Fig 2.2.2: Contents of Dataset

Alphabetic List of Variables and Attributes					
#	Variable	Type	Len	Format	Informat
11	CH_per_Learner	Num	8		
8	CH_per_Learner1	Num	8		
9	CH_per_Learner2	Num	8		
10	CH_per_Learner3	Num	8		
6	GPercentFemale	Num	8	8.	
7	GPercentMale	Num	8	8.	
2	GradPR	Num	8	BEST12.	BEST32.
1	InstitutionSize	Char	15		
4	InstitutionType	Char	50		
5	Region	Char	50		
3	VAF	Num	8	BEST12.	BEST32.

Fig 2.2.3: Datatypes and Format of the variables in the dataset

Appendix C: Box Plots for Outlier Analysis

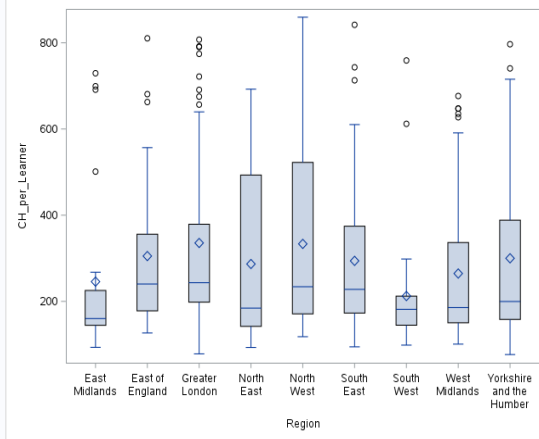


Fig 1: CH_per_Learner Vs Region

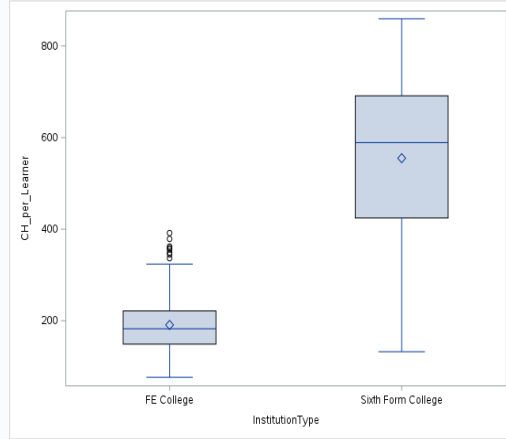


Fig 2: CH_per_Learner Vs InstitutionType

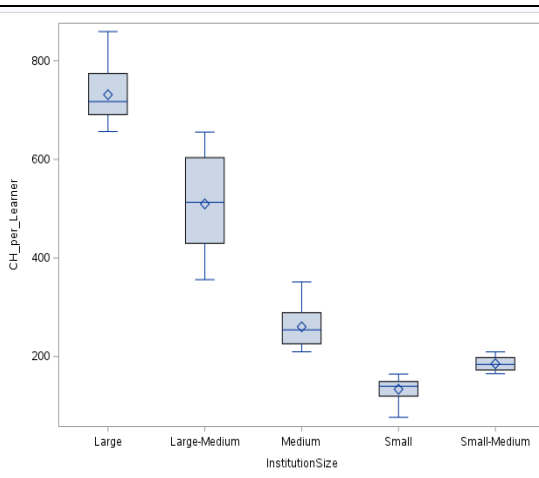


Fig 3: CH_per_Learner Vs Institution Size

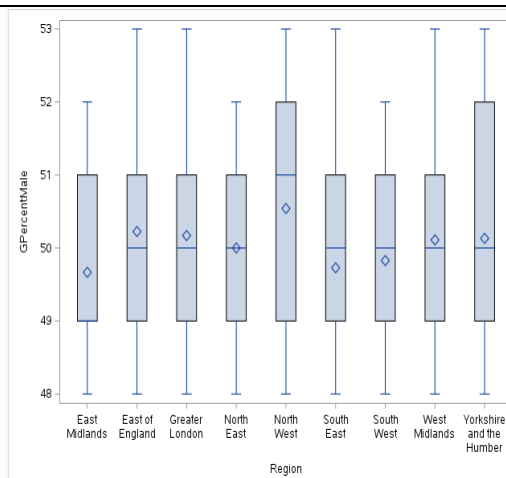


Fig 4: Male % Vs Region

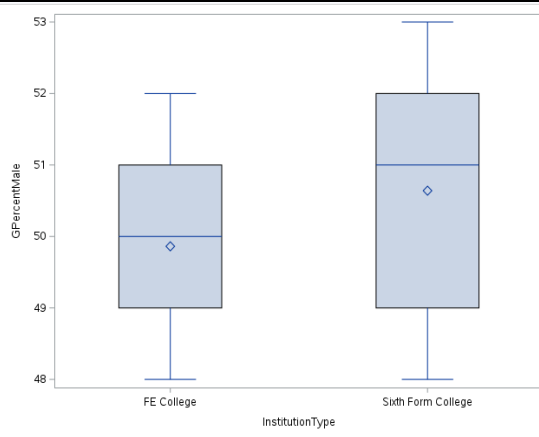


Fig 5: Male % Vs Institution Type

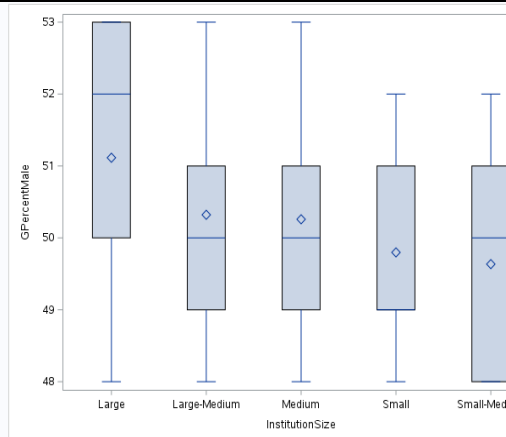


Fig 6: Male % Vs Institution Size

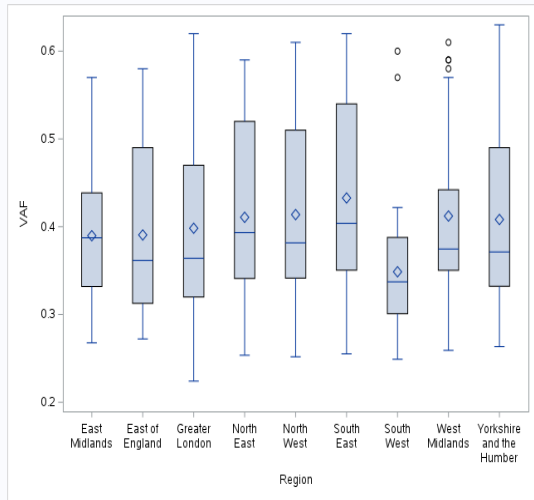


Fig 7: VAF Vs Region

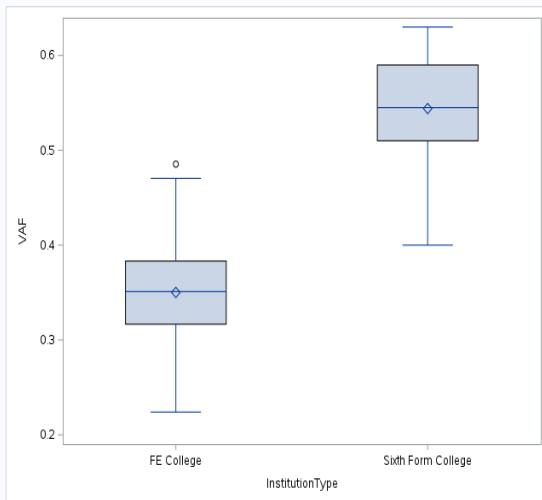


Fig 8: VAF Vs Institution Type

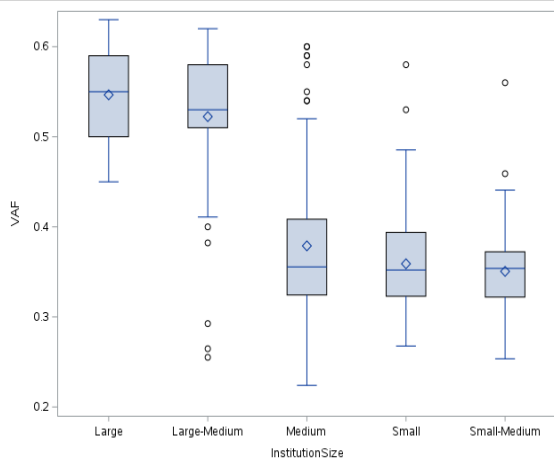


Fig 9: VAF Vs Institution Size

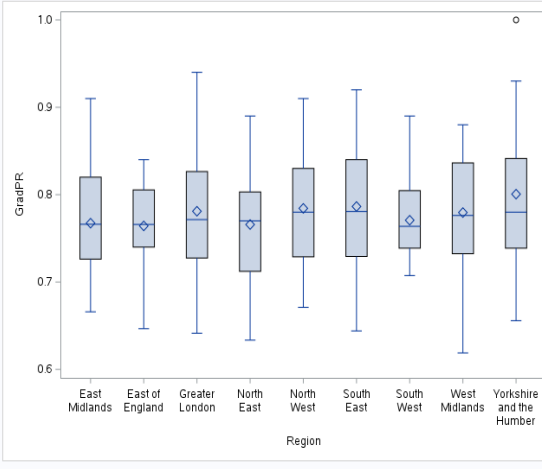


Fig 10: GradPR Vs Region

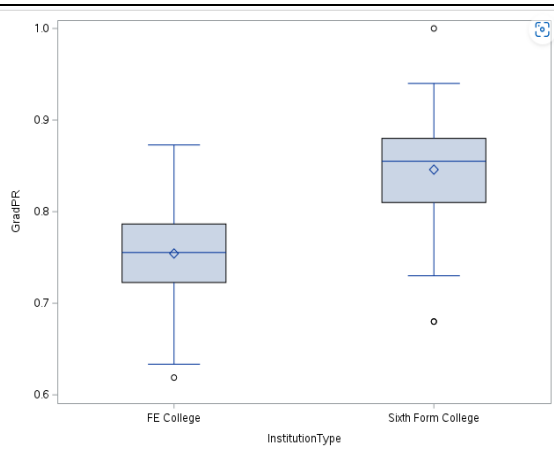


Fig 11: GradPR Vs Institution Type

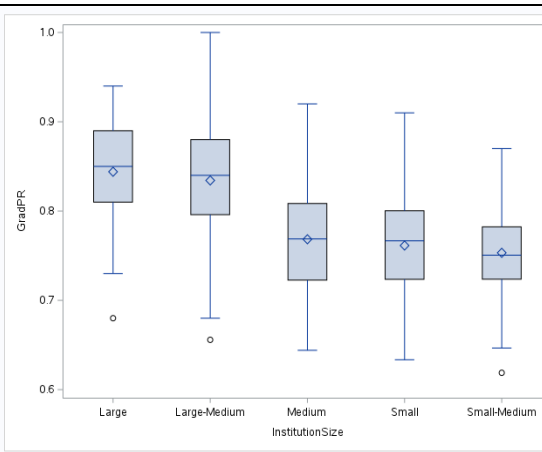


Fig 12: GradPR Vs Institution Size

Appendix D: Transformation

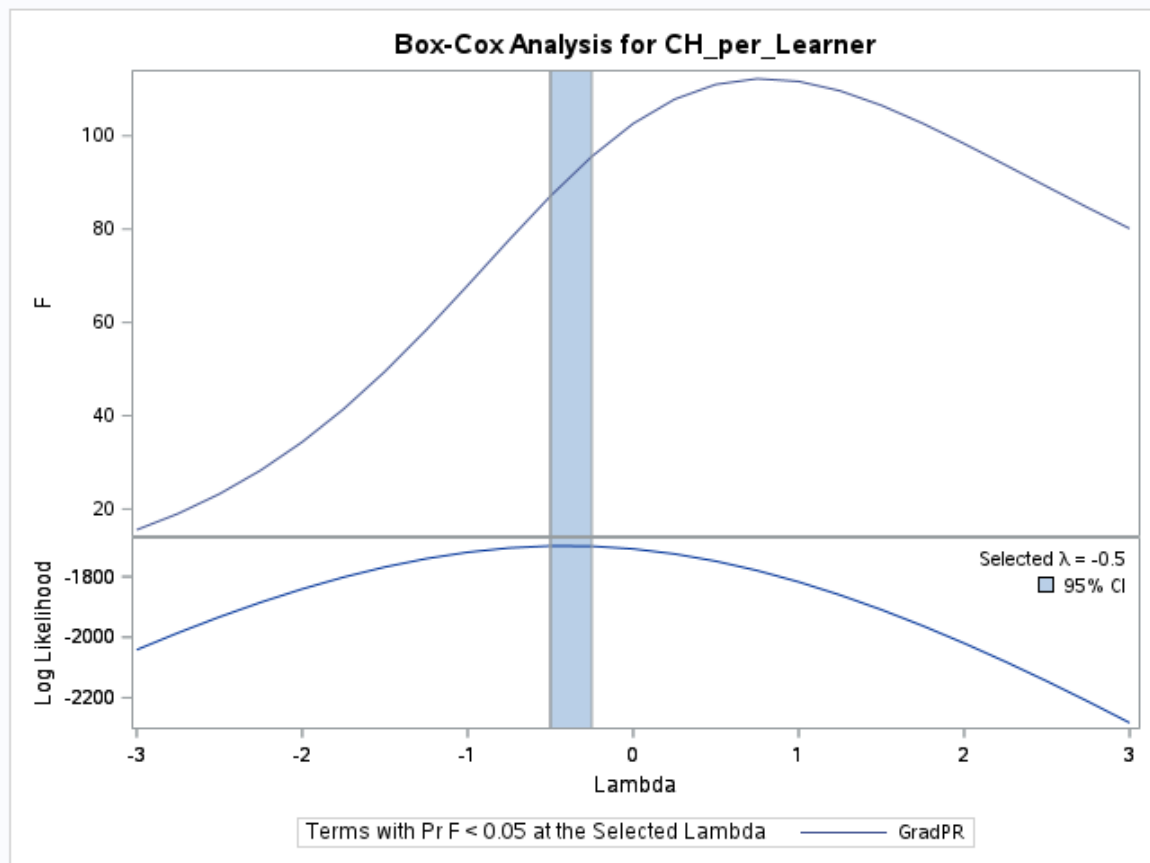


Fig 2.3.2: The box-cox analysis tells us which transformation is required on the variable. From the figure it is known that lambda value -0.5 means a reciprocal of square root of the variable i.e., CH_per_learner

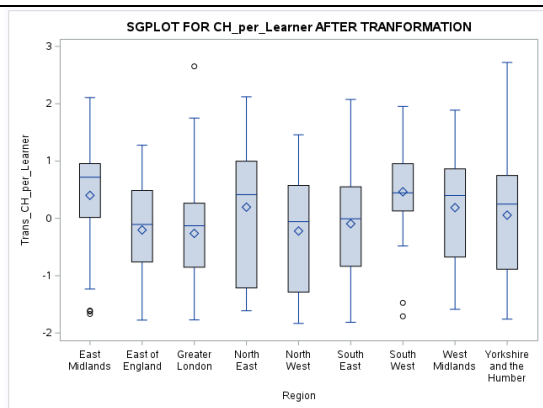


Fig 2.4.1: CH_per_learner Vs Region

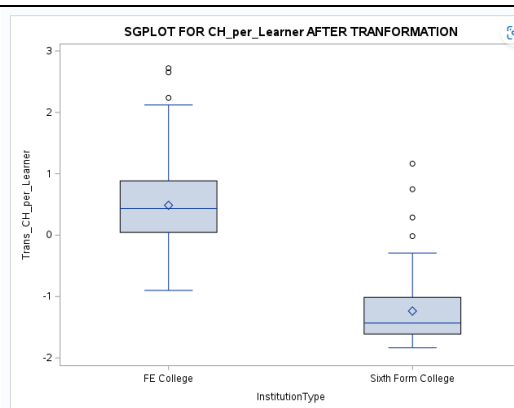


Fig 2.4.2: CH_per_learner Vs Institution Type

Appendix E: NPARIWAY

3.1.1: Graduation %, VAF, Male %, CH_per_Learner Vs Region

Statistical Modelling By Region - NPAR1WAY

The NPAR1WAY Procedure

Wilcoxon Scores (Rank Sums) for Variable GradPR Classified by Variable Region					
Region	N	Sum of Scores	Expected Under H0	Std Dev Under H0	Mean Score
East Midlands	24	3760.00	4260.00	484.022440	156.666667
East of England	31	4935.50	5502.50	544.233178	159.209677
Greater London	47	8324.00	8342.50	653.312582	177.106383
North East	21	3320.50	3727.50	454.814885	158.119048
North West	61	11304.50	10827.50	727.112538	185.319672
South East	59	10981.00	10472.50	717.529762	186.118644
South West	29	4708.50	5147.50	528.011748	162.293103
West Midlands	45	8038.00	7987.50	641.340110	178.622222
Yorkshire and the Humber	37	7465.00	6567.50	589.024554	201.756757
Average scores were used for ties.					

Kruskal-Wallis Test		
Chi-Square	DF	Pr > ChiSq
6.2387	8	0.6205

Fig 3.1.1.1: Wilcoxon Scores for GradPR

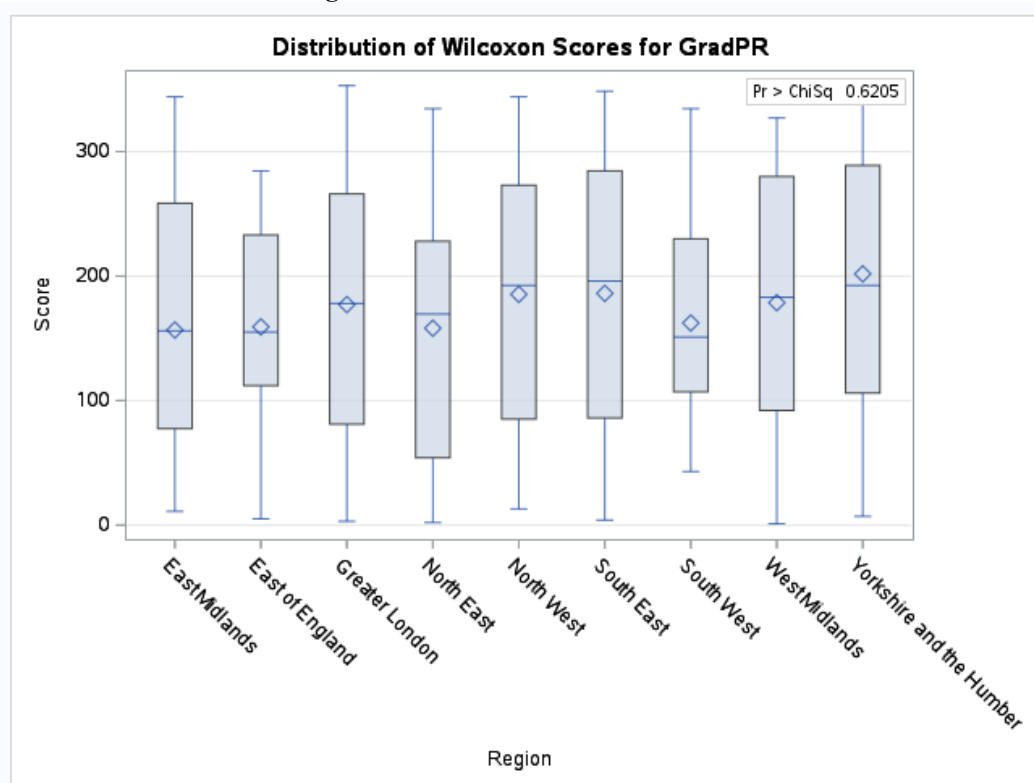


Fig 3.1.1.2: Distribution of Wilcoxon score for GRadPR

Statistical Modelling By Region - NPAR1WAY

The NPAR1WAY Procedure

Wilcoxon Scores (Rank Sums) for Variable VAF Classified by Variable Region					
Region	N	Sum of Scores	Expected Under H0	Std Dev Under H0	Mean Score
East Midlands	24	4066.00	4260.00	484.013143	169.416667
East of England	31	4911.00	5502.50	544.222724	158.419355
Greater London	47	7901.50	8342.50	653.300033	168.117021
North East	21	3851.00	3727.50	454.806149	183.380952
North West	61	11474.00	10827.50	727.098571	188.098361
South East	59	12123.00	10472.50	717.515979	205.474576
South West	29	3300.00	5147.50	528.001606	113.793103
West Midlands	45	8602.50	7987.50	641.327792	191.166667
Yorkshire and the Humber	37	6606.00	6567.50	589.013240	178.540541
Average scores were used for ties.					

Kruskal-Wallis Test		
Chi-Square	DF	Pr > ChiSq
18.8028	8	0.0160

Fig 3.1.1.3 Wilcoxon Scores for VAF

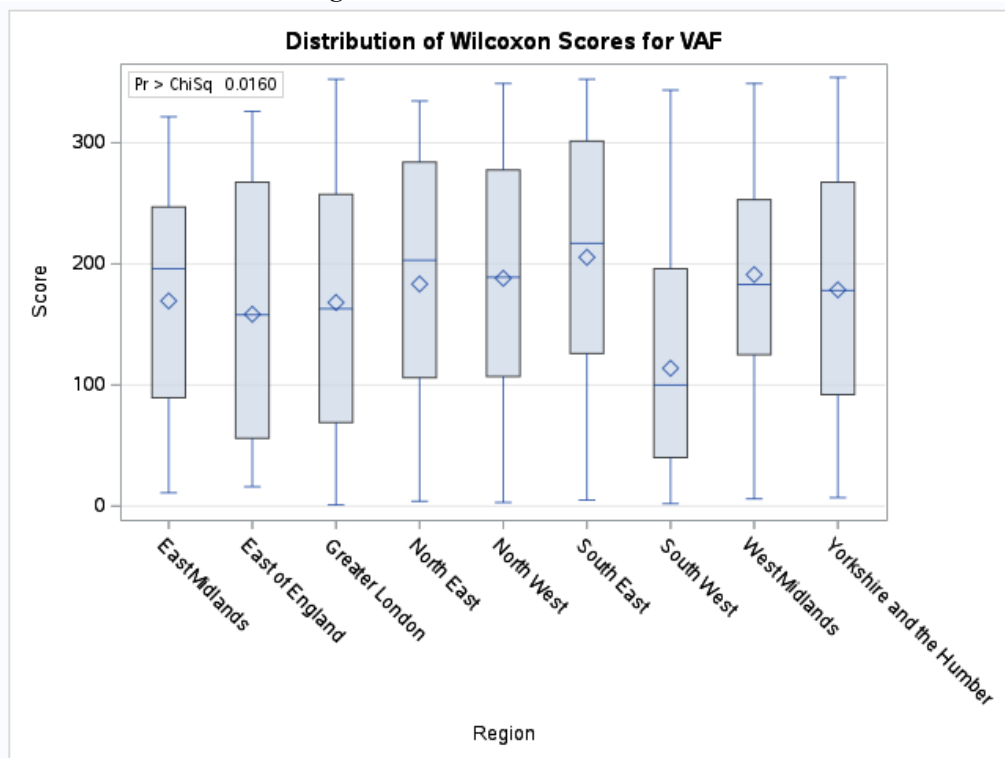


Fig 3.1.1.4: Wilcoxon Distribution of Scores for VAF

Statistical Modelling By Region - NPAR1WAY

The NPAR1WAY Procedure

Wilcoxon Scores (Rank Sums) for Variable GPercentMale Classified by Variable Region					
Region	N	Sum of Scores	Expected Under H0	Std Dev Under H0	Mean Score
East Midlands	24	3616.50	4260.00	475.441043	150.687500
East of England	31	5879.50	5502.50	534.584284	189.661290
Greater London	47	8615.50	8342.50	641.729783	183.308511
North East	21	3673.50	3727.50	446.751318	174.928571
North West	61	12639.00	10827.50	714.221314	207.196721
South East	59	9114.00	10472.50	704.808434	154.474576
South West	29	4888.50	5147.50	518.650449	161.672414
West Midlands	45	8108.00	7987.50	629.969575	180.177778
Yorkshire and the Humber	37	6500.50	6567.50	578.581538	175.689189
Average scores were used for ties.					

Kruskal-Wallis Test		
Chi-Square	DF	Pr > ChiSq
11.5156	8	0.1742

Fig 3.1.1.5: Wilcoxon Scores for Male %

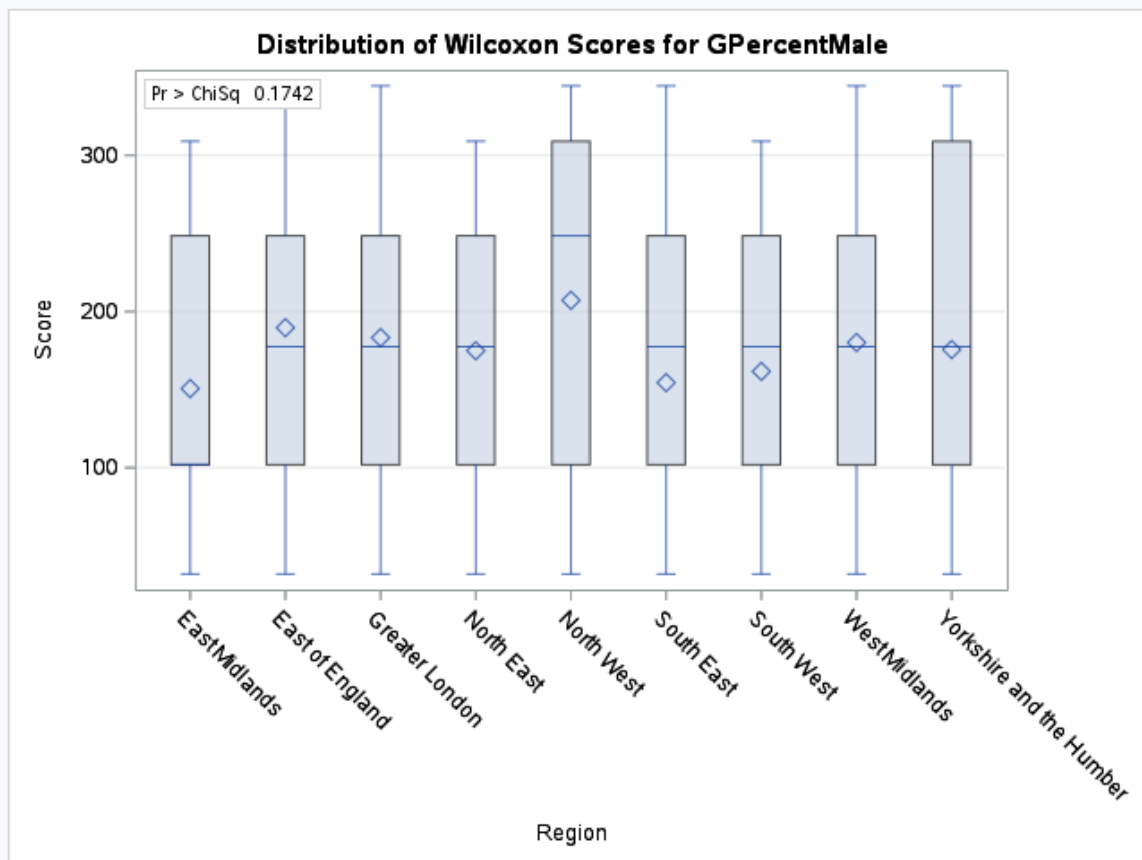


Fig 3.1.1.6: Wilcoxon Distribution of Scores for Male %

3.1.2: Graduation %, VAF, Male %, CH_per_Learner Vs InstitutionType

Statistical Modelling By InstitutionType - NPAR1WAY

The NPAR1WAY Procedure

Wilcoxon Scores (Rank Sums) for Variable GradPR Classified by Variable InstitutionType					
InstitutionType	N	Sum of Scores	Expected Under H0	Std Dev Under H0	Mean Score
Sixth Form College	100	27645.0	17750.0	866.801719	276.450000
FE College	254	35190.0	45085.0	866.801719	138.543307
Average scores were used for ties.					

Wilcoxon Two-Sample Test					
Statistic	Z	Pr > Z	Pr > Z	t Approximation	
				Pr > Z	Pr > Z
27645.00	11.4150	<.0001	<.0001	<.0001	<.0001
Z includes a continuity correction of 0.5.					

Kruskal-Wallis Test		
Chi-Square	DF	Pr > ChiSq
130.3143	1	<.0001

Fig 3.1.2.1: Wilcoxon Scores for GradPR

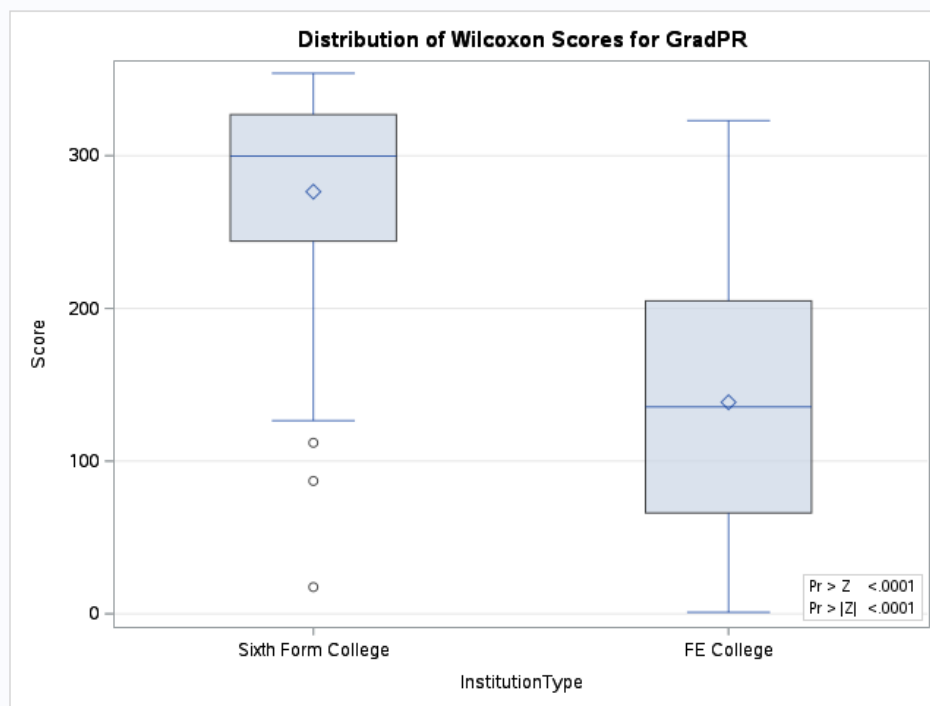


Fig 3.1.2.2: Distribution of Wilcoxon Scores for GradPR

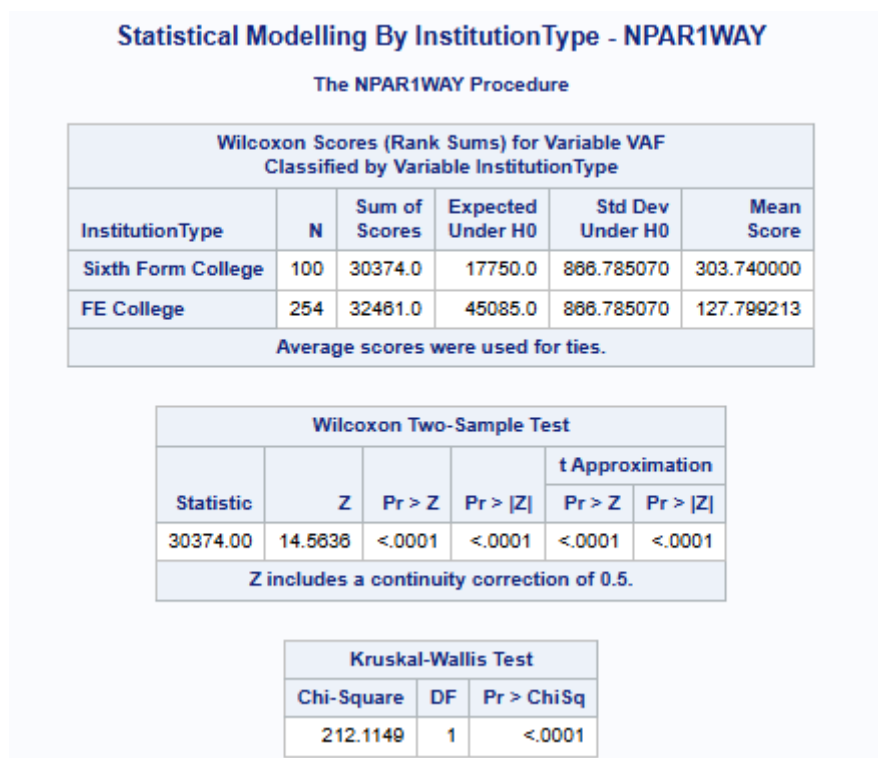


Fig 3.1.2.3: Wilcoxon Scores for VAF

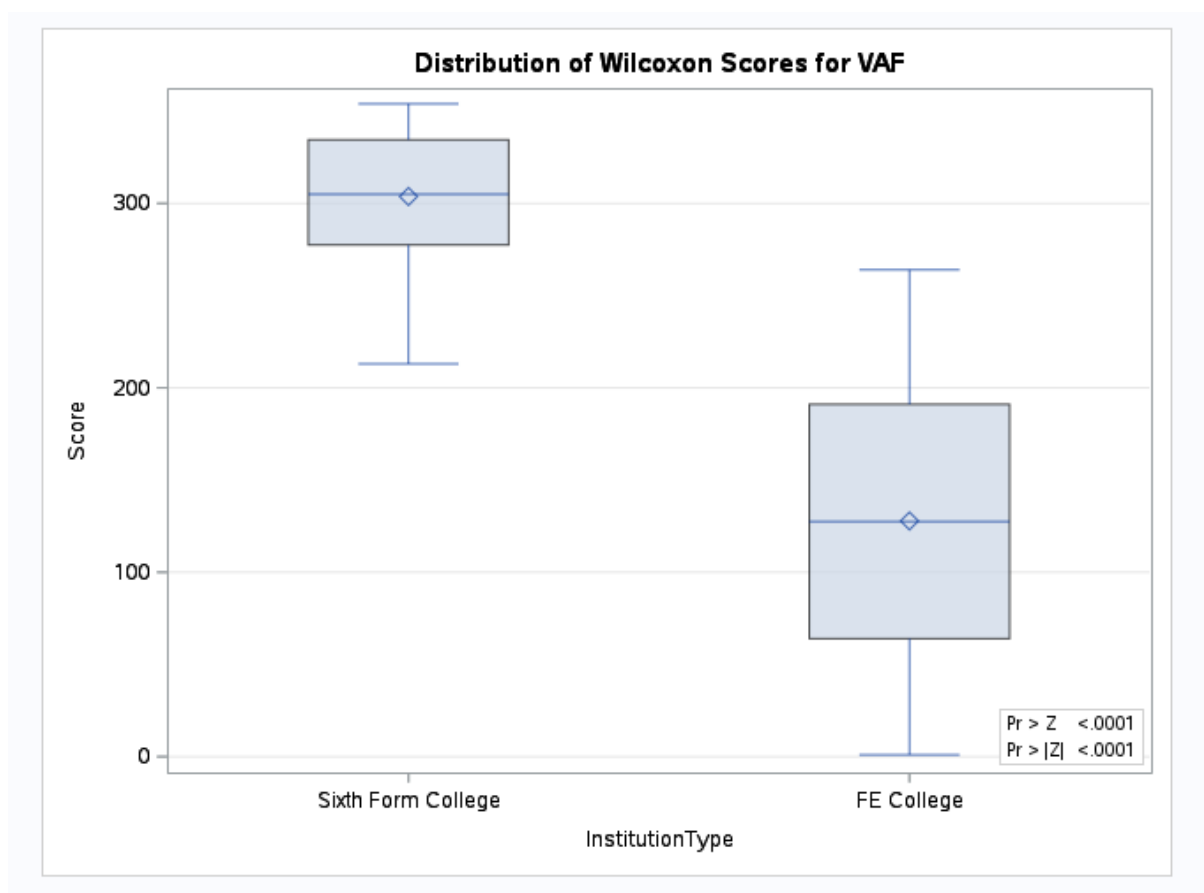


Fig 3.1.2.4: Distribution of Wilcoxon Scores for VAF

Statistical Modelling By InstitutionType - NPAR1WAY

The NPAR1WAY Procedure

Wilcoxon Scores (Rank Sums) for Variable GPercentMale Classified by Variable InstitutionType					
InstitutionType	N	Sum of Scores	Expected Under H0	Std Dev Under H0	Mean Score
Sixth Form College	100	21086.50	17750.0	851.433899	210.865000
FE College	254	41748.50	45085.0	851.433899	164.364173
Average scores were used for ties.					

Wilcoxon Two-Sample Test					
Statistic	Z	Pr > Z	Pr > Z	t Approximation	
				Pr > Z	Pr > Z
21086.50	3.9181	<.0001	<.0001	<.0001	0.0001
Z includes a continuity correction of 0.5.					

Kruskal-Wallis Test		
Chi-Square	DF	Pr > ChiSq
15.3561	1	<.0001

Fig 3.1.2.5: Wilcoxon Scores for GPercentMale

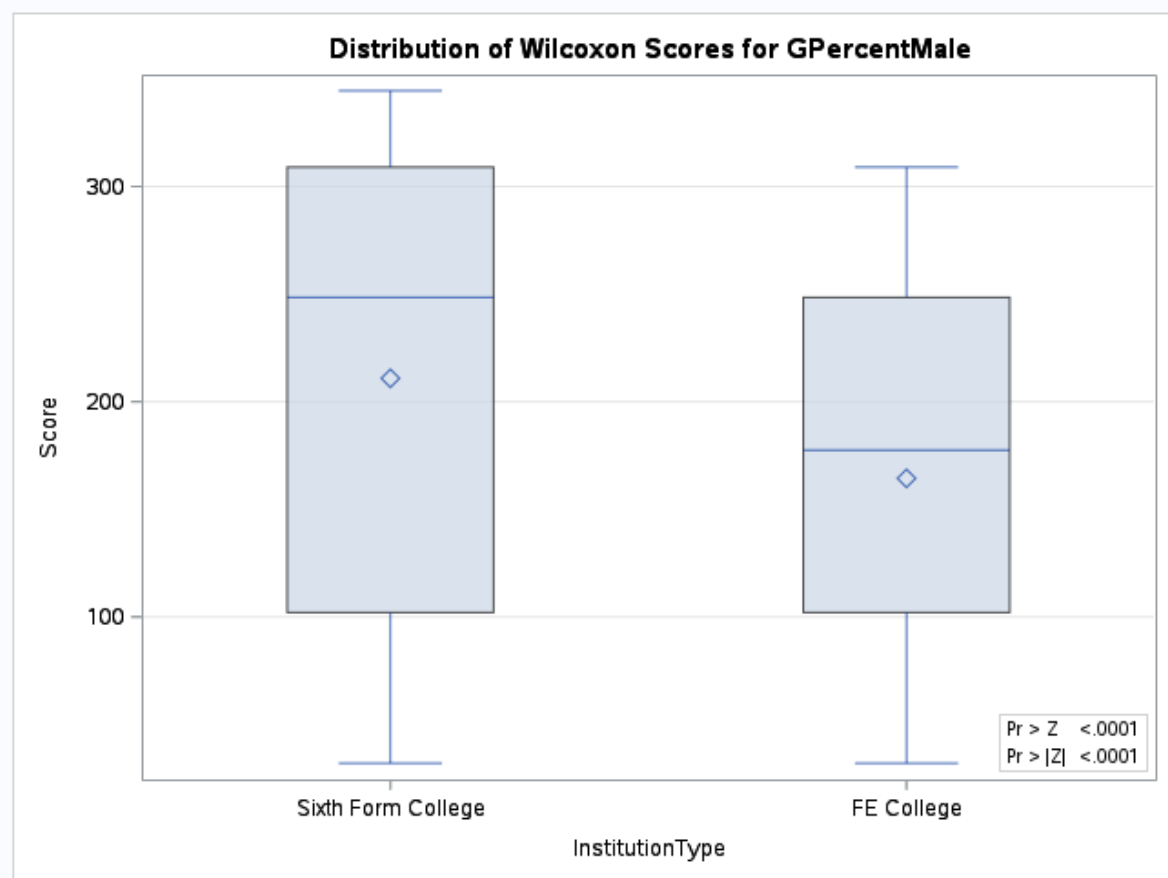


Fig 3.1.2.6: Distribution of Wilcoxon Scores for GPercentMale

Statistical Modelling By InstitutionType - NPAR1WAY

The NPAR1WAY Procedure

Wilcoxon Scores (Rank Sums) for Variable CH_per_Learner Classified by Variable InstitutionType					
InstitutionType	N	Sum of Scores	Expected Under H0	Std Dev Under H0	Mean Score
Sixth Form College	100	29702.0	17750.0	886.842931	297.020000
FE College	254	33133.0	45085.0	886.842931	130.444882

Wilcoxon Two-Sample Test					
Statistic	Z	Pr > Z	Pr > Z	t Approximation	
				Pr > Z	Pr > Z
29702.00	13.7874	<.0001	<.0001	<.0001	<.0001
Z includes a continuity correction of 0.5.					

Kruskal-Wallis Test		
Chi-Square	DF	Pr > ChiSq
190.1080	1	<.0001

Fig 3.1.2.7: Wilcoxon Scores for CH_per_Learner

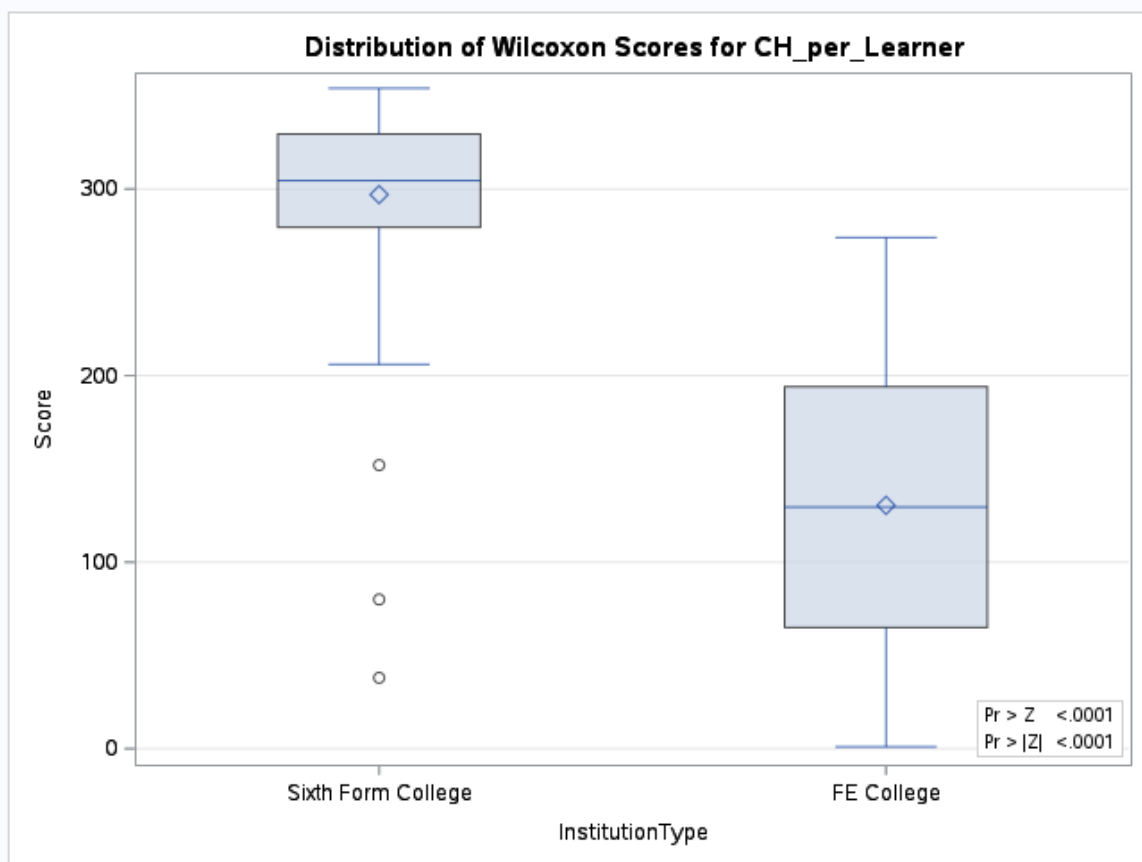


Fig 3.1.2.8: Distribution of Wilcoxon Scores for CH_per_Learner

3.1.3: Graduation %, VAF, Male %, CH_per_Learner Vs Institution Size

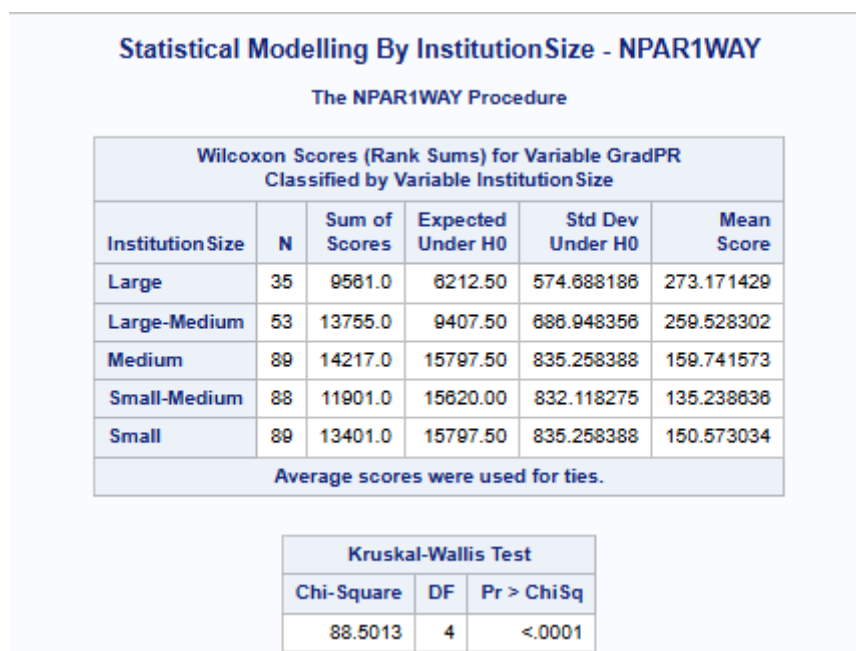


Fig 3.1.3.1: Wilcoxon Scores for GradPR

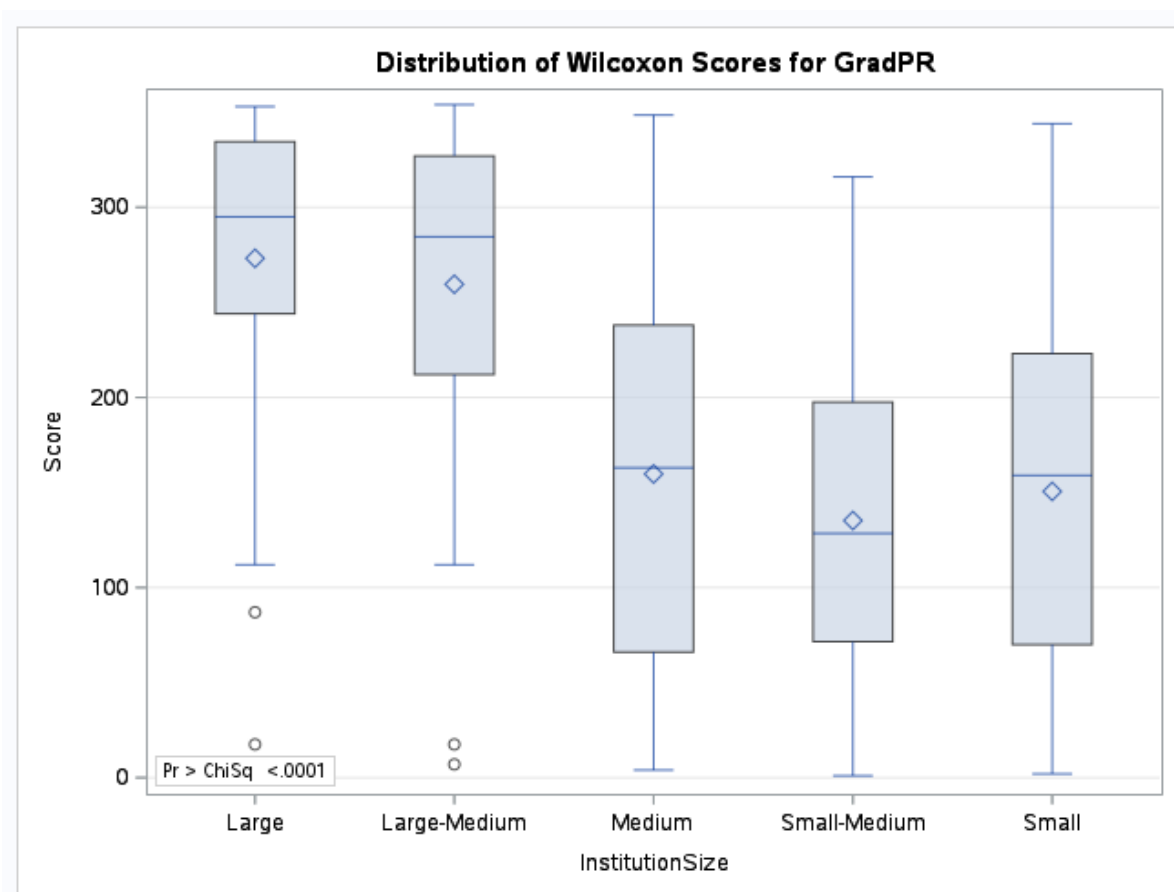


Fig 3.1.3.2: Distribution of Wilcoxon Scores for GradPR

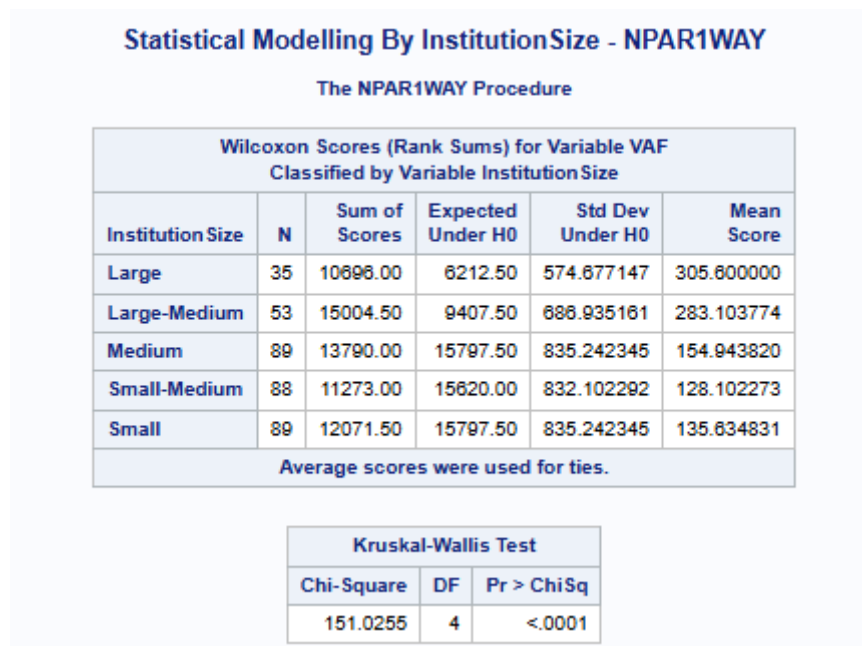


Fig 3.1.3.3: Wilcoxon Scores for VAF

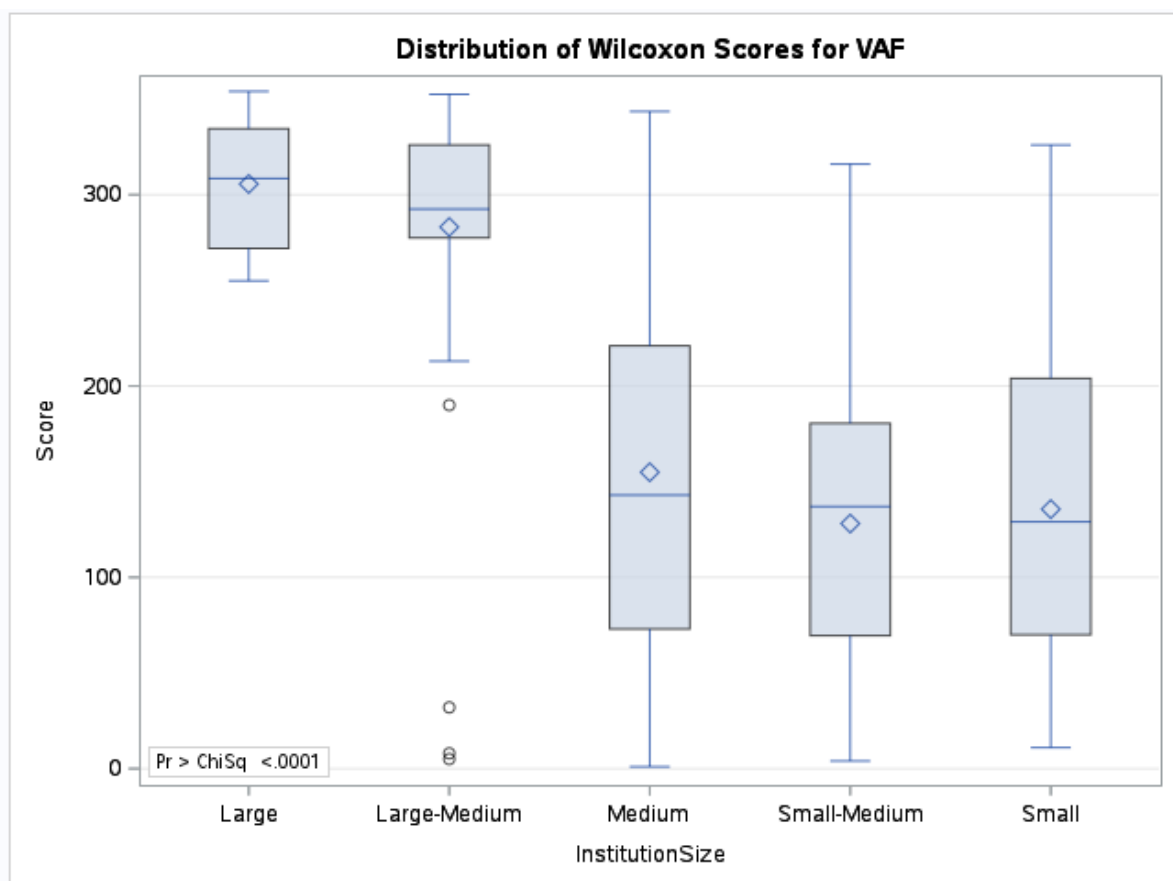


Fig 3.1.3.4: Distribution of Wilcoxon Scores for VAF

Statistical Modelling By InstitutionSize - NPAR1WAY

The NPAR1WAY Procedure

Wilcoxon Scores (Rank Sums) for Variable GPercentMale Classified by Variable InstitutionSize					
Institution Size	N	Sum of Scores	Expected Under H0	Std Dev Under H0	Mean Score
Large	35	8440.0	6212.50	564.499345	241.142857
Large-Medium	53	10113.0	9407.50	674.769216	190.811321
Medium	89	17043.0	15797.50	820.449811	191.494382
Small-Medium	88	13053.0	15620.00	817.365369	148.329545
Small	89	14186.0	15797.50	820.449811	159.393258
Average scores were used for ties.					

Kruskal-Wallis Test		
Chi-Square	DF	Pr > ChiSq
26.9852	4	<.0001

Fig 3.1.3.5: Wilcoxon Scores for GPercentMale

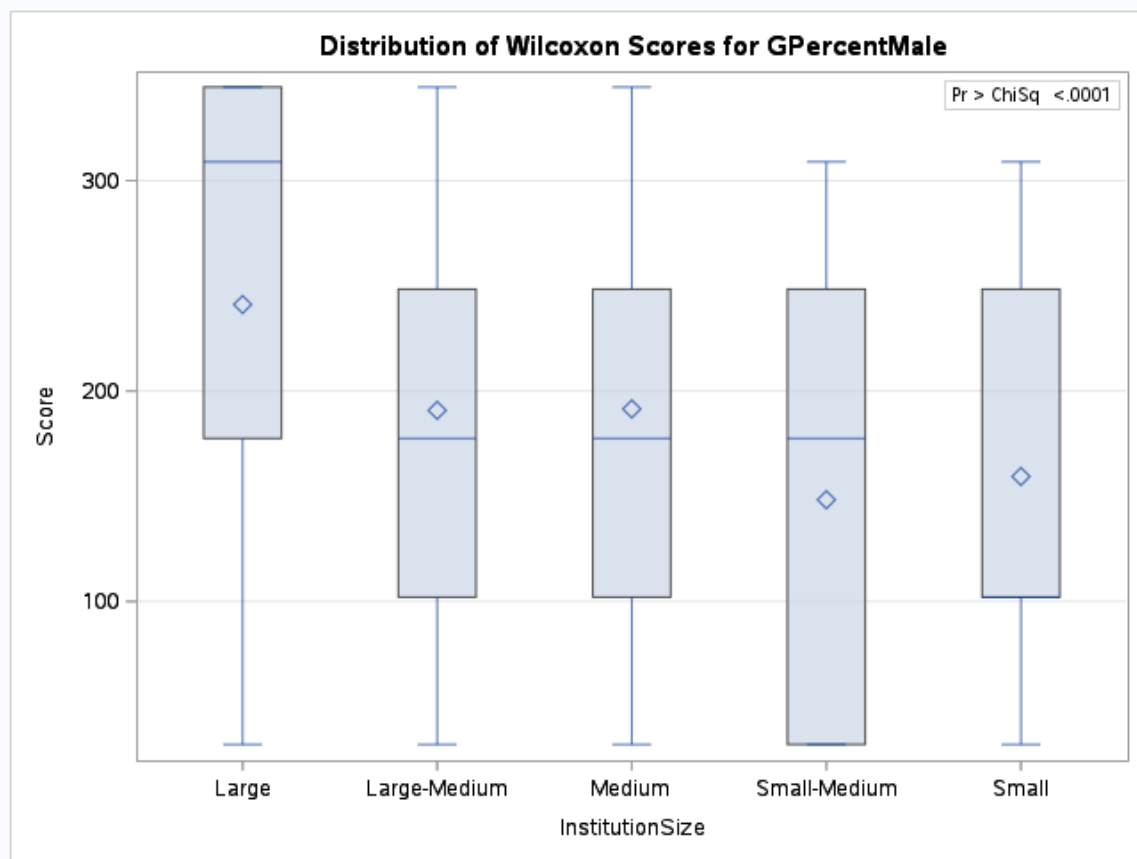


Fig 3.1.3.6: Distribution of Wilcoxon Scores for GPercentMale

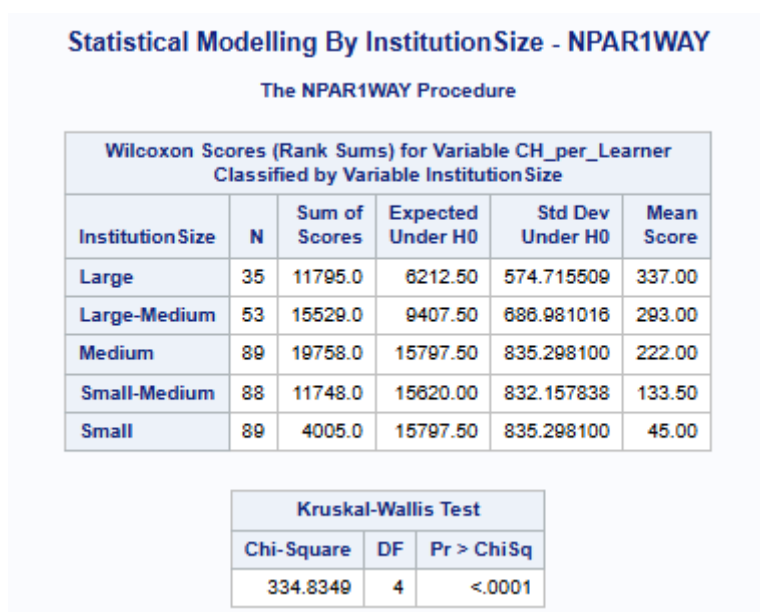


Fig 3.1.3.7: Wilcoxon Scores for CH_per_Learner

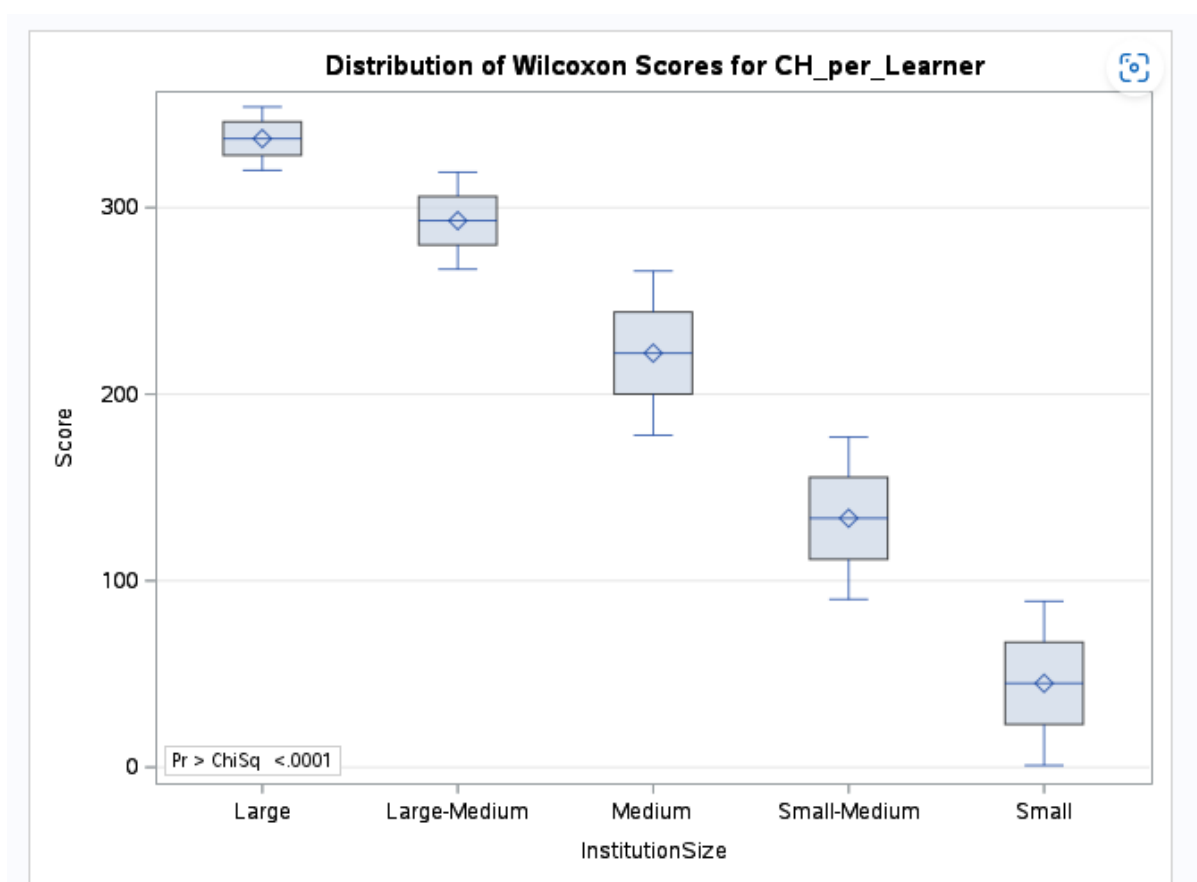


Fig 3.1.3.8: Distribution of Wilcoxon Scores for CH_per_Learner

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